# Kalman Filter Theory and Applications

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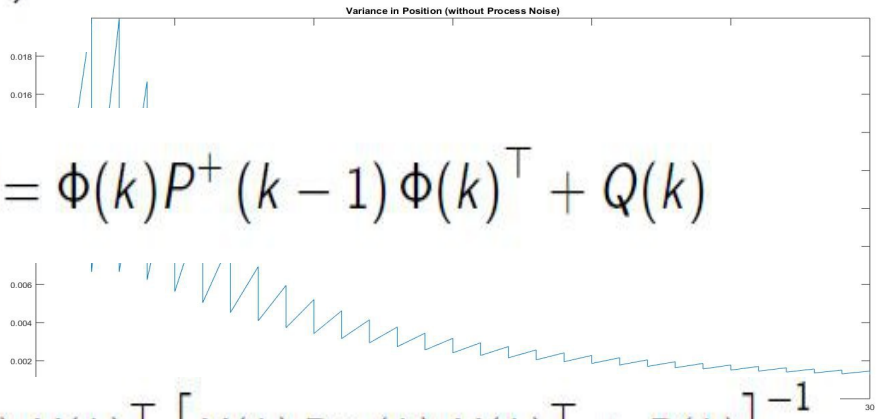
**State Extrapolation:** $\hat{x}^-(k) = \Phi(k)\hat{x}^+(k-1)$

$k-1 \rightarrow k$

$\hat{x}^+(k-1)$

**Covariance Extrapolation:** $P^-(k) = \Phi(k)P^+(k-1)\Phi(k)^\top + Q(k)$

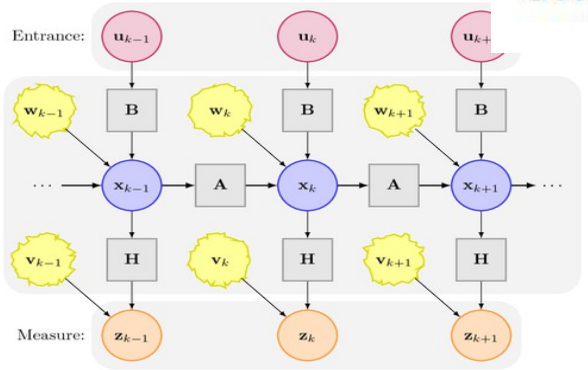
Unit Delay
$k \rightarrow k-1$




**Kalman Gain:** $K(k) = P^-(k)H(k)^\top \left[H(k)P^-(k)H(k)^\top + R(k)\right]^{-1}$

**State Update:** $\hat{x}^+(k) = \hat{x}^-(k) + K(k)\left[z(k) - H(k)\hat{x}^-(k)\right]$




**Covariance Update:** $P^+(k) = [I - K(k)H(k)]P^-(k)$

